

User Guide & Reference

Editor & Viewer

Complete operating handbook for authoring, presenting, embedding, and navigating interactive 3D Gaussian Splatting digital twins, virtual tours, and spatial narratives.

Authoring in Editor Mode

3D surface hotspot picking, multi-room portal linking, custom 360-degree backgrounds, camera view anchors, rich markdown, video/audio embeds, draft autosave, and standalone ZIP packaging.

Presenting in Viewer Mode

Clean presentation viewport, guided tour autoplay with countdown progress bar, responsive iframe embed code, vector QR quick-scan, and immersive WebXR / Mobile Gyroscope look controls.

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1. System Architecture & Fundamentals

1.1 What is Light Origin 3DGS Viewer?

Light Origin 3DGS Viewer is a state-of-the-art web application engineered for high-fidelity interactive visualization, digital twin exploration, and virtual touring of **3D Gaussian Splats (3DGS)**. Unlike traditional polygon meshes that rely on simplified triangles and baked texture maps, Gaussian Splatting reproduces volumetric radiance fields with photorealistic lighting, reflections, micro-details, and thin structures.

Built upon the modern high-performance SuperSplat and PlayCanvas rendering cores, Light Origin 3DGS Viewer executes hardware-accelerated splat sorting and rasterization natively inside web browsers utilizing modern **WebGPU** with automatic fallback to **WebGL**.

1.2 Dual-Mode Architecture: Editor vs. Viewer

Light Origin 3DGS Viewer is architected with a unified codebase featuring two dedicated operational modes:

Editor Mode (Authoring)	Viewer Mode (Presentation)
Designed for content creators, 3D artists, architects, and tour designers. Provides a comprehensive sidebar to set initial camera views, pick 3D surface points, anchor rich media hotspots, connect multi-room portals, configure 360-degree panoramas, customize branding, and export self-contained offline packages.	Designed for end-user audiences, public presentations, client showcases, and website embeds. Strips away editing panels in favor of a sleek, distraction-free view with interactive hotspots, guided autoplay, countdown progress, quick sharing, WebXR VR, and mobile gyroscope look controls.

1.3 Supported File Formats & Standards

Format	Type	Description & Operational Capabilities
<code>.sog</code>	3DGS SPLAT	SuperSplat Compressed Format. Highly compressed, streamable chunked format that delivers fast load times, lower memory consumption, and superior streaming performance over the web. (Highly Recommended).
<code>.ply</code>	3DGS SPLAT	Standard Polygon File Format containing 3D Gaussian splat parameters (positions, scale, rotation, spherical harmonics color, and opacity). Supported for direct import from NeRFstudio, Postshot, Luma, and 3DGS tools.
<code>.json</code>	PROJECT DATA	Light Origin 3DGS Viewer Scene Metadata. Serialized JSON configuration storing project title, background configurations, cameras, initial views, hotspot annotations, icon selections, rich media, and cross-scene portal targets.
<code>.zip</code>	STANDALONE BUNDLE	Self-Contained Offline Web Package generated via 1-click export. Bundles the project JSON, 3D model, JavaScript runtime, stylesheets, and index file for instant hosting anywhere without external dependencies.

1.4 Strict Naming Rule & Assets Directory Structure

MANDATORY IDENTICAL NAMING RULE:
The project's directory and all project asset files MUST have exactly the same name.
For any project (for example, named `villa`):

- The folder must be named: `./projects/villa/`
- The project metadata file must be named: `villa.json`
- The 3D model file must be named: `villa.sog` (or `villa.ply`)

Both the directory and the files must share the exact same base name (e.g. `./projects/villa/villa.json` and `./projects/villa/villa.sog`). Inconsistent naming will prevent the loader from resolving the associated 3D model.

GRAPHICAL ASSETS STORAGE: THE '/assets' DIRECTORY
All graphical assets MUST be stored inside the `/assets` folder.
This includes static background images, 360-degree equirectangular panoramas, custom logos, textures, and local illustrations. In all project definitions and configurations, reference these files via the relative path `assets/<filename>` (for example: `assets/panorama.jpg` or `assets/client_logo.png`).

2. Quick Start Guide

2.1 Launching the Application & URL Modes

The viewer is accessible through any standard modern web browser (Google Chrome, Microsoft Edge, Safari, Firefox, or Opera). The mode is governed directly by the URL parameters:

- **Editor Mode (Default):** `http://localhost:8080/index.html` or `http://domain.com/?mode=editor`
Opens the full authoring studio with the left sidebar enabled.
- **Viewer Mode:** `http://localhost:8080/index.html?mode=viewer&project=museum`
Launches clean presentation mode for project `museum` with editing tools hidden.

2.2 Loading 3D Gaussian Splats

There are two primary methods to load a 3DGS model into the Editor:

1. **Toolbar Upload:** In the left sidebar, navigate to the **3DGS Model** section and click **UPLOAD 3DGS**. Select your local `.sog` or `.ply` file.
2. **Drag & Drop:** Drag any `.sog` or `.ply` file directly from your operating system file manager and drop it anywhere over the 3D viewport canvas.

PRO-TIP: PROJECT DIRECTORY SETUP

Always organize your projects in `./projects/<project_name>/` with identically named files (e.g. `./projects/villa/villa.json` and `./projects/villa/villa.sog`). When launching `?project=villa`, Light Origin 3DGS Viewer immediately pairs and loads the scene.

2.3 Step-by-Step: Creating Your First Tour

WORKFLOW PIPELINE: FROM RAW SPLAT TO PUBLISHED VIRTUAL TOUR

1. LOAD SPLAT → 2. FRAME INITIAL VIEW → 3. SET BACKGROUND → 4. ANCHOR HOTSPOTS → 5. LINK PORTALS → 6. EXPORT / SHARE

1. **Set the Initial View:** Rotate, pan, and zoom the camera until the scene is perfectly framed. In the left sidebar, click **Set Initial View**. This viewpoint becomes the opening vantage point when users load the tour.
2. **Customize the Environment:** In the **Background** section, select between Solid Color, Linear Gradient, 2D Image, or 360-degree Panorama. Make sure all images and panoramas are located in the `/assets` folder.
3. **Add Hotspots:** Click **+ Add Hotspot**. The cursor transforms into a crosshair. Click anywhere on the 3D geometry surface. A glowing marker is anchored at that exact 3D coordinate.
4. **Configure Content:** In the **Properties** panel, choose an icon from the built-in library, enter a Title, format a Rich Text description using Markdown, and optionally paste an image URL or YouTube video link.
5. **Bind Camera Angles:** Rotate your camera to the desired viewing angle for that hotspot, and click **Update Camera**. When users click this hotspot in Viewer mode, the camera will smoothly glide to this exact vantage point.
6. **Share & Export:** Click **Share & Embed Tour** for responsive iframe embed snippets and mobile QR codes, or click **Export Standalone (.zip)** for a completely offline self-hosted web package.

3. Editor Mode — Complete Functional Reference

3.1 Workspace Layout & Header

The Editor interface is structured into two primary zones:

- **Left Sidebar (Authoring Panel):** A collapsible vertical control deck containing project configuration cards, environment pickers, hotspot management lists, properties inspectors, and project actions.
- **3D Viewport:** Full-window interactive WebGPU canvas rendering the splat cloud, spatial hotspot markers, real-time lighting, and background skybox.
- **Toast Notification System:** Non-intrusive floating notifications at the top of the screen that confirm actions (e.g. *"Initial view set"*, *"Loading 3DGS: 45%"*, *"Project saved"*).

3.2 Project Identity & Title Editing

At the top of the sidebar under the Light Origin 3DGS Viewer header, the project name is displayed. Click on the project name to activate the inline editor:

- Type the new name (e.g. *"Grand Architectural Hall"*).
- Press **Enter** or click outside to commit the name.
- Press **Esc** to cancel edits.
- This name is automatically used in the browser tab title, metadata exports, and generated zip package filenames.

3.3 Environment & Background Configurations

Light Origin 3DGS Viewer provides 5 distinct background rendering modes accessible via the **Type** dropdown:

Mode	Controls	Behavior & Best Use Cases
Transparent	None	Clears the background buffer entirely. Ideal when embedding the 3D model inside existing web pages, allowing the underlying website background or hero section to show through.
Solid Color	Hex Color Picker	Renders a uniform solid backdrop. Clean, professional look that focuses complete attention on the 3D splat object.
Gradient	Color 1, Color 2, Angle Slider (0°–360°)	Generates a sleek two-tone linear gradient. The dynamic angle slider allows directional lighting effects that match the dominant lighting in your 3D capture.
2D Image	Select Image Button	Loads a static backdrop image (JPG/PNG/WebP stored in <code>/assets</code>) positioned behind the 3D canvas with <code>cover</code> aspect ratio scaling.
360° Panorama	Select Panorama Button, Rotation Slider (-180° to +180°)	Mounts an equirectangular 360-degree spherical skybox (stored in <code>/assets</code>). As the user orbits the 3D camera, the panorama rotates synchronously. The horizontal rotation slider permits aligning the skybox lighting with the 3DGS splats.

3.4 Tour Playback Settings & Step Dwell Times

Configures how the tour behaves upon opening in Viewer mode:

- **Autospin on load:** When checked, the camera begins a slow, elegant 360-degree ambient orbit around the scene center upon loading. Pauses automatically whenever user interaction begins.
- **Autoplay tour on load:** When checked, Viewer mode automatically starts the guided hotspot sequence 800ms after scene initialization.
- **Tour Duration (Dwell Time):** Dropdown selector configuring how many seconds the camera pauses at each hotspot before transitioning to the next: `3 Seconds` (rapid reel), `5 Seconds (Default)`, `8 Seconds`, `10 Seconds`, and `15 Seconds` (narration/study).

3.5 Viewer Logo & Custom Brand Upload

Empowers studios, agencies, and enterprises to whitelabel the presentation experience:

- **Show logo in Viewer:** Master toggle to display or hide the floating brand mark in the lower-left corner of the Viewer.

- **Light Origin (Default):** Displays the official Light Origin logo mark, automatically switching between light and dark variants based on the active theme.
- **Custom Logo Upload:** Allows uploading custom client branding (PNG, SVG, WebP, JPG stored in `/assets`). Features interactive thumbnail preview, filename display, and a reset button. Logos are serialized as base64 data URLs for seamless portability.

3.6 Initial View & Camera 0 Calibration

The initial view defines the exact camera coordinates, pitch, yaw, target focus point, and field of view (FOV) when a visitor first opens the project.

1. Navigate freely around the scene to frame the desired perspective.
2. Position the camera to your optimal hero angle.
3. Click the **Set Initial View** button. A confirmation toast ("*Initial view set*") confirms `camera-0` is recorded.

RESERVED CAMERA-0 PROTECTION:

Light Origin 3DGS Viewer reserves `camera-0` exclusively for the scene's initial entry view. If any hotspot is assigned to `camera-0`, the system automatically migrates that hotspot to an independent camera ID (e.g. `camera-1`) during import, ensuring the opening view is never corrupted or overwritten.

3.7 3DGS Model Upload & Management

The **3DGS Model** section controls active splat assets in memory:

- **Upload 3DGS:** Launches a system file picker filtered for `.sog` and `.ply` files with live upload progress percentage.
- **Delete 3DGS:** Unloads the active model from WebGPU/WebGL memory, resets the canvas, and frees GPU resources.
- **Model Swapping:** Uploading a new model automatically unloads the previous model to prevent GPU memory leaks.

3.8 Hotspot Authoring, Surface Picking & Ordering

Hotspots are 3D spatial points anchored directly to surfaces of the Gaussian splat cloud:

Placing a Hotspot

Click **+ Add Hotspot**. The button turns active and the mouse cursor transforms into a precision crosshair. Click anywhere on the 3D model. Raycasting computes the exact surface intersection coordinate and creates the marker.

Reordering & Deleting

Hotspots appear in the sidebar list. Use the **Up** and **Down** buttons to reorder annotations. The guided tour navigates through hotspots in this exact top-to-bottom sequence. Click delete to remove.

Moving Existing Hotspots

To reposition a hotspot without losing attached descriptions or media: select the hotspot, press the **M** key, and click the new target location on the 3D surface.

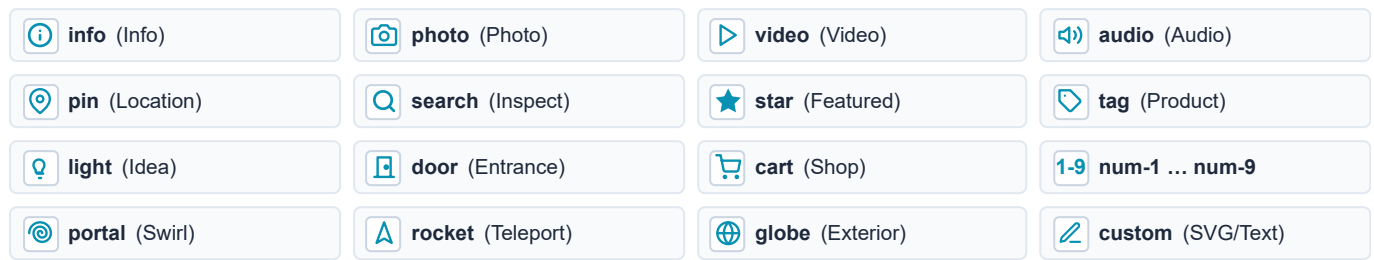
3.9 Hotspot Properties, Markdown & Media Attachments

Selecting any hotspot opens its comprehensive **Properties** card:

- **Title & Color:** Edit the headline (up to 30 characters) and pick a custom dot accent color from the palette picker.
- **Rich Text Description (Markdown):** Supports GitHub-flavored markdown syntax (`**bold**`, `*italic*`, `[Link Title] (url)`, `# Headings`, `- Lists`, and ``code``).
- **Media Attachments:** Select between Image, Video, or Audio:
 - **Image:** URL to JPG, PNG, WebP image (or local path under `/assets`). Renders as a responsive banner frame inside the card.
 - **Video:** Paste YouTube URLs (auto-converts to `youtube-nocookie.com/embed`), Vimeo URLs, or direct MP4/WebM files.
 - **Audio:** URL to MP3 or WAV audio clip (or local audio in `/assets`). Embeds an audio player for voiceover narrations or ambient sound effects.
- **Call-to-Action (CTA) Button:** Add an interactive button (e.g. "*Reserve Tickets*" or "*Buy Now*") with target URL.
- **Camera Controls:** **Go To Camera** (flies to the bound viewpoint) and **Update Camera** (overwrites with current view).

3.10 Icon Library & Custom Glyphs

Light Origin 3DGS Viewer includes an integrated vector SVG icon library with 20+ specialized glyphs rendered with scalable vectors:



CUSTOM EMOJI & SVG SUPPORT:

Selecting the **Custom** icon reveals an input field. You can input custom SVG XML markup, or enter up to 3 text characters. The system automatically formats the glyph with proportional scaling.

3.11 Cross-Scene Portals (Virtual Touring)

Portals represent gateways connecting distinct 3DGS scenes into an interconnected virtual multi-room facility, estate, or museum.

- **Adding a Portal:** Click **+ Add Portal** and click the 3D mesh (e.g. on a doorway or staircase).
- **Visual Distinctions:** Portals feature a distinctive glowing cyan swirl icon surrounded by an animated pulsing halo.
- **Target Project URL:** Enter the relative or absolute path to the target project JSON (e.g. `./projects/garden/garden.json`). Remember that the target folder and its files must have the same name.
- **Isolated Navigation:** Portals appear in a separate **Portals** list in the Editor and are intentionally excluded from the linear step-by-step hotspot tour.
- **Cinematic Transition Screen:** When clicked, the screen smoothly cross-fades into a dark glassmorphism overlay with an animated spinning swirl, preventing asset flicker while the new 3DGS model loads.
- **Browser History Synchronization:** Entering a portal invokes `history.pushState` with the updated project URL. Users can seamlessly use their browser's native **Back** and **Forward** buttons to retrace their steps.

3.12 Project Actions, File Saving & Draft Autosave

- **New Project:** Resets all cameras, hotspots, and environment parameters back to a clean default state.
- **Save Project:** Generates and downloads the complete `<project_name>.json` file to your computer.
- **Open Project:** Opens a file dialog to load existing project JSON files.
- **Draft Autosave (LocalStorage):** Every edit made in the sidebar is automatically cached in browser localStorage. If the browser tab is accidentally refreshed or closed, the editor restores your working session on next launch.

3.13 Standalone Offline Export (.zip Package)

Light Origin 3DGS Viewer provides an automated **1-Click Standalone Export** feature via the **EXPORT STANDALONE (.ZIP)** button:

What the ZIP Contains

1. Tailored standalone `index.html`
2. Active 3DGS model file (`.sog` or `.ply`)
3. Complete project configuration (`project.json`)
4. Full engine JavaScript runtime & modules
5. High-resolution styles, icons & logos
6. Settings and renderer configurations

Deployment Flexibility

The extracted folder is 100% self-contained. It can be hosted on any static web host (AWS S3, Netlify, Vercel, GitHub Pages, Apache, NGINX) or opened directly via a local static web server without requiring internet connectivity or external APIs.

3.14 Share & Embed Modal Generator

Clicking **Share & Embed Tour** opens an interactive modal generator containing ready-to-use distribution tools:

Feature	Functionality & Capabilities
Direct Shareable Link	Generates the exact URL pointing directly to the presentation Viewer. Includes a 1-click COPY LINK button.
Mobile QR Code	Dynamically generates a sharp vector QR code encoding the direct viewer URL. Point any iOS or Android phone camera at the screen to test the tour immediately. Includes a Download QR (.svg) button for print media and marketing flyers.
Embed Customization	Interactive checkboxes that dynamically update the generated code: Responsive 16:9 Aspect Ratio — Wraps iframe in a responsive CSS container. Autospin Camera on Load — Appends <code>&autospin=true</code> parameter. Autoplay Guided Tour on Load — Appends <code>&autoplay=true</code> parameter. Minimal UI Mode — Appends <code>&noui=true</code> to conceal UI overlays.
HTML Iframe Code	Provides standard responsive <code><iframe></code> HTML markup with pre-configured permissions for fullscreen, spatial tracking, accelerometer, and gyroscope. Includes a 1-click COPY CODE button.

3.15 Editor Color Themes

Light Origin 3DGS Viewer features 4 professionally calibrated color themes accessible via the swatches in the sidebar:

Theme	Palette Tone	Visual Character & Aesthetic
Dark (Default)	Charcoal & Teal (#22C7B8)	High-contrast modern dark mode optimized for creative studios and long authoring sessions. Reduces eye fatigue.
Light	Clean White & Emerald (#11998E)	Bright, crisp gallery aesthetic with high legibility. Automatically swaps the header and viewer logos to dark variants.
Midnight	Deep Indigo & Neon Violet (#9B87FF)	Sci-fi, futuristic dark mode with soft purples and luminous accents. Excellent for tech and automotive presentations.
Earth	Warm Umber & Terracotta (#E8893A)	Rich organic warmth with copper and clay tones. Perfectly complements cultural heritage, museums, and real estate.

3.16 Editor Keyboard Shortcuts Cheat Sheet

Key	Action	Description
Delete	Delete Selected Hotspot	Removes the currently selected hotspot and its camera from the project.
F2	Rename Hotspot	Instantly focuses the title input in the sidebar for rapid inline renaming.
M	Move Hotspot Mode	Enters relocation mode. Click any new point on the 3D surface to reposition.
[	Previous Hotspot	Selects and navigates to the preceding hotspot in the tour list.
]	Next Hotspot	Selects and navigates to the subsequent hotspot in the tour list.
Esc	Cancel Operation	Aborts active hotspot placement mode, move mode, or dismisses open dialogs.
F	Frame Scene	Centers and frames the entire 3D model within the current viewport.
R	Reset Camera	Resets the camera back to the default overview perspective.

4. Viewer Mode — Complete Functional Reference

4.1 Presentation Interface & Clean Layout

In **Viewer Mode** (accessed via `?mode=viewer` or standalone export packages), the application transforms into an ultra-clean presentation stage:

- All authoring panels, sidebars, and property editors are completely removed.
- The 3D splat environment fills 100% of the display window.
- Branding logo rests elegantly in the lower-left corner.
- Quick-action floating toolbar sits discretely in the upper-right corner.
- Spatial 3D markers float unobtrusively over the Gaussian splat geometry.

4.2 Spatial Hotspot Markers & Smooth Camera Fly-To

Each hotspot in the 3D environment is rendered as a spatial circular billboard marker:

- **Dynamic Occlusion & Scaling:** Markers maintain legible sizing regardless of camera distance and dynamically track their 3D anchor points during camera moves.
- **Hover States:** Hovering a marker expands a soft glow halo and displays the title label.
- **Smooth Interpolated Fly-To:** Clicking any marker triggers a smooth mathematical camera transition that glides the viewer to the exact position, pitch, yaw, and field of view bound to that hotspot.

4.3 Floating Tour Card, Step Navigation & Rich Media

Upon arriving at a hotspot (or selecting it via navigation controls), the **Tour Card** appears smoothly at the bottom-center of the screen:

Media Player Frame

If the hotspot includes an image, YouTube video, Vimeo stream, MP4 file, or audio narration, it renders at the top of the card in a 16:9 responsive frame with full media playback controls.

Card Content & CTA

Displays the hotspot title, full formatted markdown text (bold, italic, links, lists), and the custom action button (CTA) that links directly to external resources or ecommerce checkouts.

Tour Navigation Bar

Integrated at the bottom of the card:

-  **Previous Button:** Transitions to the prior hotspot in the sequence.
-  **Next Button:** Advances to the subsequent hotspot in the sequence.
- **Step Counter** ( / ): Indicates current step number out of total hotspots in the tour.
- **Card Dismissal:** Clicking anywhere on open 3D space (or pressing ) dismisses the card. Built-in drag threshold detection ensures that rotating or panning the camera does not accidentally dismiss the card.

4.4 Guided Tour Autoplay & Smart Interaction Pause

The guided tour engine provides an automated hands-free walkthrough experience:

- **Play / Pause Button:** Toggles automatic sequence playback.
- **Countdown Progress Bar:** A sleek animated bar spans across the bottom of the card, visibly counting down the dwell time (e.g. 5 seconds) remaining until the next camera transition.
- **Smart Interaction Pause:** If a visitor manually touches, rotates, pans, or zooms the camera while autoplay is running, the autoplay engine pauses immediately, granting the user full manual control without unexpected transitions. When the user stops interacting, autoplay resumes automatically.

4.5 Top-Right Quick Action Toolbar

The floating action bar in the top-right provides instant utility buttons:

-  **Share & Embed:** Opens the sharing modal for instant links, iframe snippets, and mobile QR codes.
-  **VR / Motion Gyroscope:** Toggles spatial VR viewing or mobile phone tilt-to-look controls.
-  **Fullscreen:** Toggles native browser borderless fullscreen mode.

4.6 WebXR Immersive VR & Smartphone Gyroscope Mode

Light Origin 3DGS Viewer provides dual spatial immersion capabilities:

Immersive WebXR VR Mode

On devices equipped with WebXR headsets (Meta Quest 2/3/Pro, Apple Vision Pro, Pico 4, HTC Vive), clicking the VR button launches true stereoscopic 6-DOF immersive virtual reality inside the headset.

Mobile Gyroscope Mode

On smartphones (iPhone & Android), clicking the button activates the internal motion gyroscope. Tilting and rotating your physical phone smoothly steers the 3D camera, providing a realistic "magic window" into the space.

IOS 13+ DEVICEORIENTATION PERMISSION:

On Apple iOS devices, mobile browsers require explicit user permission to read gyroscope sensors. Light Origin 3DGS Viewer automatically invokes the native `DeviceOrientationEvent.requestPermission()` prompt upon the first tap.

4.7 Complete URL Query Parameters Dictionary

All viewer behaviors and settings can be controlled via URL parameters for seamless CMS and website integrations:

Parameter	Values	Functional Purpose
<code>mode</code>	<code>viewer</code> <code>editor</code>	Toggles presentation mode (<code>viewer</code>) or authoring studio (<code>editor</code>).
<code>project</code>	<code><slug></code>	Specifies project name/folder to load (e.g. <code>?project=museum</code>).
<code>autospin</code>	<code>true</code> <code>false</code>	Forces camera ambient rotation to start automatically on load.
<code>autoplay</code>	<code>true</code> <code>false</code>	Forces guided tour sequence autoplay to commence immediately.
<code>noui</code>	<code>true</code>	Hides all bottom buttons and floating toolbar for pure minimal embeds.
<code>theme</code>	<code>dark</code> , <code>light</code> , <code>midnight</code> , <code>earth</code>	Overrides project theme with specified visual style preset.
<code>logo</code>	<code>none</code> <code>false</code>	Hides the floating logo mark in the lower-left corner.
<code>webgl</code>	<code>flag</code>	Forces WebGL rendering engine instead of modern WebGPU.
<code>budget</code>	<code><number></code>	Sets maximum splat rendering limit (e.g. <code>budget=1000000</code>) for lower-end hardware.

5. Deployment, Performance & Best Practices

5.1 Model Compression & Performance Optimization

To ensure fluid 60 FPS performance and rapid streaming across mobile connections:

- **Use .SOG wherever possible:** SuperSplat compressed (`.sog`) files are up to 70% smaller than raw `.ply` files and support progressive chunk streaming.
- **Clean Floating Splats:** Before exporting your 3DGS capture, crop ground boundaries and remove floaters/sky artifacts in SuperSplat to minimize unnecessary rasterization load.
- **Optimize Media URLs:** Keep images under 1500px resolution and leverage YouTube / Vimeo embeds for video rather than massive uncompressed MP4 files.
- **Local Assets Organization:** Store all project images, skyboxes, and audio narration files directly in the `/assets` folder to prevent broken references.

5.2 Web Server MIME Types & HTTPS Requirements

Requirement	Standard	Configuration Notes
HTTPS / SSL	Mandatory for Sensors	Modern browsers (iOS Safari, Android Chrome) block DeviceOrientation, Gyroscope, and WebXR APIs on unencrypted HTTP connections. Always host production tours over HTTPS.
SOG MIME Type	<code>application/octet-stream</code>	Configure your web server (NGINX, Apache, IIS) to serve <code>.sog</code> and <code>.ply</code> files as binary streams (<code>application/octet-stream</code>).
CORS Headers	<code>Access-Control-Allow-Origin: *</code>	When loading 3DGS models or project JSON files across separate subdomains, ensure cross-origin resource sharing headers are configured.

5.3 Troubleshooting Guide

Issue Encountered	Probable Cause	Recommended Resolution
Black Screen on Launch	WebGPU unsupported by browser or older graphics driver.	Add <code>?webgl</code> to the URL to force the WebGL compatibility renderer, or update graphics drivers.
Mobile Gyroscope Fails	Insecure HTTP origin or permission denied.	Ensure page is served via <code>https://</code> and click "Allow" when the iOS motion sensor prompt appears.
YouTube Video Blocked	Video owner disabled embedding.	Ensure the YouTube video privacy is set to <i>Public</i> or <i>Unlisted</i> with "Allow embedding" enabled in YouTube Studio.
Hotspot Displaced	Model was re-centered or replaced.	Select the hotspot, press <code>M</code> , and click the 3D surface to re-anchor it to the new geometry.
3D Model Not Loading	Project folder and file names do not match.	Ensure the directory in <code>./projects/<name>/</code> and the files inside (<code><name>.json</code> and <code><name>.sog</code>) share the exact same name.